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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 2: Frequency (Co-channel) Reuse: Cluster Size, Capacity and S/I

Program Code:

```
import numpy as np
import matplotlib.pyplot as plt
N = np.array([3, 4, 7, 9, 12])
gamma, total = 4, 336
Q = np.sqrt(3 * N)            # Co-channel reuse ratio D/R
ch = total / N                # Channels per cell
si_db = 10 * np.log10(Q**gamma / 6)  # S/I in dB
# Display output
print("----- Frequency Reuse Results -----")
print("N    Q    ch/cell  S/I(dB)")
for n, q, c, s in zip(N, Q, ch, si_db):
    print(f"{n:<4d} {q:5.2f} {c:7.1f} {s:7.2f}")
print("\nResult: Frequency reuse parameters calculated successfully.")
# Plot graph
plt.plot(N, si_db, "o-", label="S/I Ratio")
plt.axhline(18, linestyle="--", color="red",
          label="18 dB (AMPS requirement)")
plt.xlabel("Cluster Size N")
plt.ylabel("S/I (dB)")
```

```
plt.title("Cluster Size vs S/I Ratio")

plt.legend()

plt.grid(True)

plt.show()
```

```
>
= RESTART: D:/knowledge_Engineering/mobile computing/program02.py
----- Frequency Reuse Results -----
N     Q     ch/cell   S/I(dB)
3      3.00    112.0    11.30
4      3.46     84.0    13.80
7      4.58     48.0    18.66
9      5.20     37.3    20.85
12     6.00     28.0    23.34

Result: Frequency reuse parameters calculated successfully.
```

